

Activity 1.1: Design Blueprints

Turkeys in Pens

0. Software.

We will be using R and Rstudio in this course. To keep things simple for this first day, you can use PositCloud if you do not have R and RStudio set up already for a different Class.

Alternatively, for **this activity only** you can use any software that you like to generate the plots and conduct the analysis.

Scenario

We are interested in examining the effect of dietary protein on growth in Standard Bronze farm-raised turkeys. There are 24 turkeys available for the study.

The turkeys are housed 3 to a pen, and each pen has a single feed trough, meaning all turkeys within a pen receive the same diet.

Two diets are studied:

- Low-protein diet: 15% protein
- High-protein diet: 30% protein

Diets are assigned at random to pens, with 4 pens receiving each diet (8 pens total).

The turkeys are fed their assigned diet for 6 weeks. The primary outcome measured on each turkey is average daily weight gain (ADG), in lbs/day.

1. Sketch a blueprint for this study.

2. Describe the study structure

- Treatment structure:
- Design structure:

3. Pen-Level Data

- Open the file `turkey_pen_data.csv`.
- What does each row in this data file represent?
- How many observations are there per diet? Why?
- Create a plot of pen-level ADG on the y-axis and Diet on the x-axis (Hint: each diet should have 4 dots).

- e. Create a summary table showing the mean and standard deviation of pen-level ADG for each diet.

Diet	Mean pen ADG	SD pen ADG
Low (15%)		
High (30%)		

4. Animal-Level Data

- f. Open the file `turkey_animal_data.csv`.
- g. What does each row in this data file represent? How many rows are there? Why?
- h. Create a plot of animal-level ADG on the y-axis and Diet on the x-axis.
- i. How many measurement/sampling units are there per diet in this plot?
- j. Create a summary table showing the mean and standard deviation of ADG for each diet.

Diet	Mean ADG	SD ADG
Low (15%)		
High (30%)		

- k. Compare the animal-level means and standard deviations to the pen-level results. What do you notice?

5. Comparing Analyses

Now consider a two-sample t-test comparing the low-protein and high-protein diets.

Pen-level analysis

- l. Using `turkey_pen_data.csv`, conduct a two-sample independent t-test (assume unequal variances).

Report:

- Difference in mean ADG
- Standard error
- t-statistic, degrees of freedom, and p-value

Animal-level analysis

- m. Using `turkey_animal_data.csv`, conduct the same two-sample independent t-test (assume unequal variances).

Report:

- Difference in mean ADG
- Standard error

- t-statistic, degrees of freedom, and p-value

6. Interpretation

n. Which analysis is *correct* for answering the research question? Explain.

Your explanation should address:

- What the experimental unit is
- Why one analysis artificially inflates the sample size
- Why this matters for statistical inference

7. Submit on Gradescope

When you are finished, submit your work to **Gradescope**:

1. Make sure your submission includes:
 - Your study blueprint sketch
 - Your written answers for parts (b)–(n)
 - Any requested tables/values from JMP or R
2. Combine everything into **one PDF**.
 - If you wrote on paper: take clear photos and save/print to PDF.
 - If you typed responses: export to PDF.
3. Go to **Canvas > Gradescope > Activity 1.1: Design Blueprints** and upload your PDF.
4. If Gradescope asks you to assign pages to questions, do so before submitting.